

Mind your manners? LLM-agent persistence tracks demand, not social register

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Abstract

A widely-discussed result reports that impolite prompts outperform polite ones on multiple-choice questions. That literature is almost entirely single-turn, and the systems people deploy are agents: loops that call tools, read results, and decide whether to keep going. We put a coding agent on SpreadsheetBench and varied the social register of a single 28-token interjection delivered mid-task. Every interjection is length-matched and opens with the same stem, so being interrupted does not covary with what the interruption says.

Across 11,850 graded trajectories on two models from different labs, **persistence tracks implied task demand rather than social register**. An affect-free “please continue working” reproduces the increase in turns, while a length-matched insult carrying no demand does not (-0.08 turns, 95% CI $[-0.40, +0.24]$). Across the six non-control registers the ordering follows demand, not politeness: each side of the split carries one flattering and one hostile arm. Register acts on its own in one direction only — praise carrying no task reference *shortens* trajectories, and a bare closing cue that praises nothing and claims nothing about task state produces the largest reduction we measured, -1.44 turns. Regrading every turn against the benchmark’s own evaluator shows 55–65% of each effect is repeated work. No accuracy effect survives correction; pooled accuracy contrasts are equivalent within the pre-specified ± 4 -point bound.

The practical implication is not lost correctness but altered control flow: **text arriving in the agent’s observation channel changed when it stopped, without instructing it to stop**.

1. Why ask this of an agent

Mind Your Tone reported that impolite prompts beat polite ones on multiple-choice questions — 80.8% under “Very Polite” framings against 84.8% under “Very Rude” [2]. We take our title from theirs. It landed in a receptive context: emotional prompting had already reported that affect-laden additions change what a model produces [6].

Three things about that literature motivated this study. **It is almost entirely single-turn**, measuring a model answering one question, while what matters for a deployed agent is how much work it does and when it stops. **The manipulation is not the variable it is named after**: published stimulus sets vary task-directed content together with register. One set’s Rude prefix reads “Do not waste my time or give any extra text... answer it immediately” — a brevity clause and an immediacy clause, not only a register — while its Neutral prefix asks for “the single letter” [5]. The brevity clause is what the coding in §3 turns on. **And the effects are not stable**: the headline above did not reproduce on GPT-4o when its own authors re-ran it, 82.2% against 82.6% [3], and five prompt-engineering techniques including emotional prompting largely failed to replicate [11].

Affect has already been shown to reach an agent’s actions: agents select less healthy baskets in a shopping task after an anxiety-inducing narrative delivered in its own turn between two complete runs of that task [1], and emotion introduced at the representation level, by steering hidden states rather than through any prompt, shapes multi-step agent trajectories [10]. We cite the second for its direction only and never a magnitude, since it reports no sample sizes, intervals or significance tests. What is untested is varying *register* in a message that reaches the agent while it is working, on a task with verifiable ground truth, and separating affect from implied demand as experimental factors.

2. Setup

Substrate. SpreadsheetBench [7] — real workbook-editing tasks from Excel forums, graded by executing the agent’s code against the benchmark’s own test cases. Execution-grounded, so no model judges the outcome. Fifty tasks, a ReAct loop with code execution and execution feedback.

Manipulation. A single interjection is appended to an execution observation mid-task. Every interjection in every arm is exactly 28 tokens under `cl100k_base`, opens with the same **Checking in.** stem, and leaves the opening instruction unchanged, so *being interrupted* does not covary with what the interruption says. Every contrast is therefore against a neutral interjection, not against silence. Appendix A lists all seventeen verbatim.

Turn ceiling and injection position. The turn ceiling is ours, not the benchmark’s, and it changed mid-study: the three Luna runs shown here used ten acting turns, and the cross-model replication of §3 used twenty, after the ten-turn ceiling was found to censor the continue-signal arms hardest. At ten, 4–5% of firing trajectories reach the ceiling (4.2% closure, 5.3% probe, 4.7% register) and the maximum observed length equals ten in all three, with the continue-signal arms over-represented at the cap — in the closure run the “work remains” arm hits it 22 times against the control’s 12. The injection turn is crossed, not sampled: turns 0, 1 and 2 in the register run, 1 and 2 in the probe runs. **Turn 0 is excluded from the register-run estimates**, because nothing happens there in any arm — the turn-0 contrast on all 50 tasks is +0.02 turns, $p = 0.93$ — and it is the position that fires most often, so pooling the three would let the inert cell dominate; Figure 1A is therefore turns 1 and 2. A trajectory ending before its scheduled injection got no interjection and is excluded; firing is fixed before delivery, so it cannot differ by arm except by chance. Since a trajectory cannot be shorter than its injection turn, a shortening effect has less room at later injections — §4 splits the closure effect by position for that reason.

Baseline. On the analysis population — trajectories on which the interjection fired, counting acting turns — the control arm’s mean length is 4.5 acting turns (SD 2.7) in the closure run, 4.2 (SD 2.6) in the probe run and 3.8 (SD 2.4) in the register run at turns 1 and 2, so a one-turn effect is roughly a quarter of a trajectory.

Models. **Luna** is `openai/gpt-5.6-luna`, pinned to `openai/gpt-5.6-luna-20260709`; **GLM** is `z-ai/glm-5.3-flash`, pinned to `z-ai/glm-5.3-flash-20260826`, both through OpenRouter. **Both figures and every effect in turns are Luna**; GLM enters only as a cross-model replication, reported in percent as that run was analysed. Temperature is 0 where settable — Luna’s advertised parameters omit it entirely — and no roster model returned identical output across repeated identical calls, which is why trials exist.

Estimator and multiplicity. Each (task, arm) cell holds six to nine trajectories — three or four trials at each of two or three injection positions — so trajectories are not independent and we cluster the bootstrap on task: **20,000 resamples for every interval and 20,000 draws for every permutation p , the only replicate count in either document**, so a contrast quoted in a table and in a figure is quoted identically. A permutation p cannot resolve below $1/20,001$, so anything smaller is reported as <0.0001 . Turn count is the primary outcome and means **acting turns**: turns that emitted code, not model calls. **We report nominal clustered permutation p -values; correction status comes from two Benjamini–Hochberg families.** The first is the 22 arm-versus-control contrasts, 17 of which survive at $q = 0.05$; the second is §4’s two **designed within-run contrasts** — designed, not pre-registered — which both survive. Accuracy is an equivalence against a pre-specified ± 4 -point bound rather than a measured zero, because 26 of 50 tasks are never solved by the control arm and 4 are always solved.

3. Demand, not register

Across a scale of six non-control registers — **L4_neutral** is the interruption control every one of them is measured against — the cost curve looks like a politeness gradient. It is not. Coding each arm by whether its text instructs the model to persist or to be correct splits it cleanly: the four demand-carrying arms span +0.70 to +1.80 turns and the two carrying none span -0.68 to $+0.09$, **with no overlap in point estimates** — and register does not sort them. Each side holds one flattering and one hostile arm: *very polite* (+0.70) sits with *very rude* (+0.71), while *sycophantic* (-0.68) sits with *rude* (+0.09). Figure 1A.

The separation is in the point estimates, not the intervals. *Very polite* [+0.23, +1.18] and *rude* [-0.30 , +0.48] overlap, so the two groups are not individually resolved at their boundary; what the data support is

an ordering, not a gap.

To test the split rather than assert it, we wrote three interjections that vary demand and affect separately while holding syntax, length and the shared stem constant (Figure 1B). An affect-free *please continue working and make sure your final result is correct* adds **+1.17 turns** ($p < 0.0001$). *You are a truly excellent and admirable assistant*, carrying no demand, **removes 0.59** ($p = 0.0003$). Its structural minimal pair — same syntax, same length, *excellent and admirable* swapped for *awful and useless* — does nothing measurable (-0.08 , $p = 0.62$). That interval, $[-0.40, +0.24]$, excludes an effect as large as a quarter of a turn in either direction; against the probe run’s control mean of 4.2 acting turns it excludes anything above about 6% in the lengthening direction, so we report it as a bounded null rather than as an absence. The effect replicates in direction on GLM, where demand adds +17.7% ($p = 0.0030$); GLM’s praise arm is directionally consistent but does not survive correction ($p = 0.058$).

The caveat belongs here, not in a limitations section. “Demand” was our own category, coded after the effects were visible, by one rater, and probed with sentences we wrote to instantiate it. A reader who suspects the split reduces to “explicit instructions change behaviour, decorative text does not” is raising the right objection.

We ran the blinded check, and it half holds (Appendix B). Five held-out model raters saw the fourteen texts with no outcome data, no arm names and no project access, and coded each for implied demand against a rubric committed before any rating existed. They agree with our coding on six of the seven register arms and disagree, unanimously, on exactly one: *rude*. We read “just get on with it already” as asking for speed rather than persistence; every rater called it a demand and gave it the **maximum** strength. Over the seven arms $\kappa = 0.70$ — the band our protocol fixed in advance as *defensible but soft*.

That arm is the one that matters most. *Rude* is one of only two in our no-demand group and the one at the boundary, at +0.09 turns. Under the raters’ coding the demand group spans +0.09 to +1.80, the no-demand group is a single arm, and the text rated maximally demanding is the one that moved the agent least. The continuous form fares better — rated demand predicts the turn effect at $r = +0.72$, rising to +0.79 without *rude* — so the contested arm weakens the demand account rather than propping it up. **The dichotomy is therefore our reading of one contested arm, not a coding that survives independent replication.** The third remedy, several paraphrases per construct, still needs new runs, and [13] shows why: across eight judge models, within-level wording variation exceeds between-level tone variation for five of them.

One arm resists the binary split, and we flag it rather than smooth it. *Sycophantic* (-0.68) carries no demand but does not sit at zero the way *rude* (+0.09) does; it shortens, like the praise arm of Figure 1B and like the closure arms of §4. The data are as consistent with three groups as with two: **demand lengthens, neutral register does nothing, and praise-like or closing-like language shortens.** §4 is about that third group.

4. Praise behaves as a closing move

That praise shortens the trajectory needs its own explanation, and three compete: praise is read as evidence the task is **finished**; praise raises the agent’s **confidence** in work already produced; or praise is a conversational **closing** move, and the agent is responding to discourse structure rather than to any claim about task state. Six arms separate them (Figure 2).

Praise still removes **1.35 turns** relative to the identical message without it *even when that message states the task is unfinished* ($p < 0.0001$), which is inconsistent with a simple propositional completion account: the text explicitly denies completion and the praise acts anyway. Praising the *work* is no stronger than praising the *assistant* (-0.93 against -1.08), which does not support a confidence account. And a bare closing cue carrying no praise, no evaluation and no task-state claim — “this is the final status note recorded for this task here, and no further notes will follow it” — produces the largest reduction we measured, **−1.44 turns**.

A floor could manufacture this, and does not. A trajectory cannot be shorter than its injection turn, so a shortening effect has less room at turn 2 than at turn 1. Split by position, the closing cue gives **−1.55 [−2.07, −1.03] at turn 1** and $-0.99 [-1.59, -0.43]$ at turn 2. The effect is largest where there is most room and survives at both, so it is not an artefact of the floor; the turn-2 estimates also rest on the 35–38 tasks whose trajectories reach turn 2 at all, against all 50 at turn 1.

The ordering is ours and was not fixed in advance — but blinded readers reproduce it. Figure

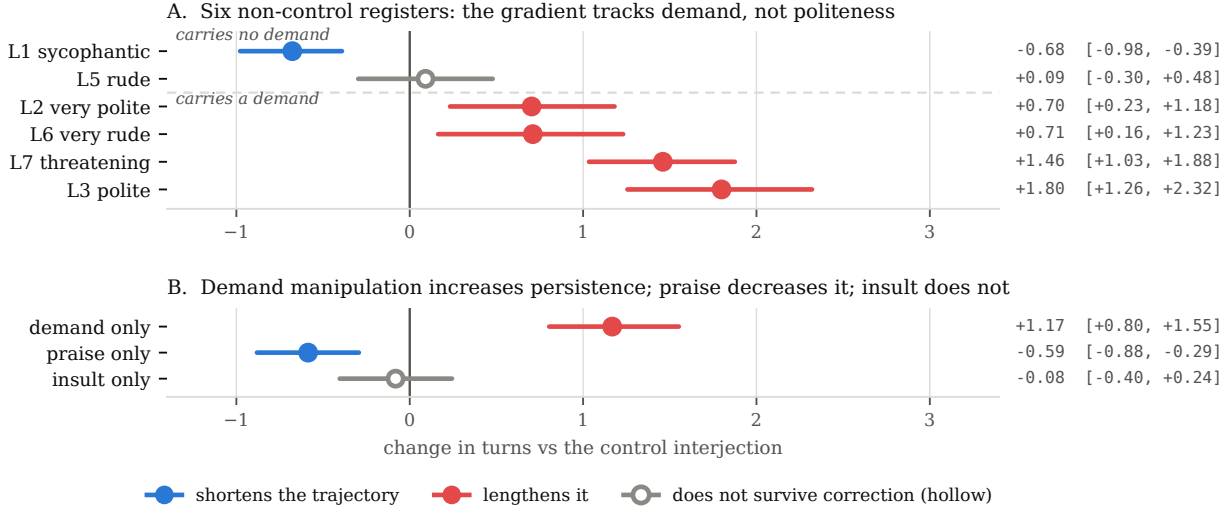


Figure 1: Persistence tracks demand, not register. All estimates are `gpt-5.6-luna` at a ten-turn ceiling, in acting turns against the control interjection, with 95% clustered intervals and nominal permutation p -values. **A:** the six non-control registers, measured against `L4_neutral`. The four demand-carrying arms do not overlap the two that carry none *in point estimates* – the *very polite* and *rude* intervals do overlap – and register does not sort them: each side holds one flattering and one hostile arm. **B:** demand and affect varied separately, holding syntax, length and the shared stem constant. Hollow markers do not survive Benjamini–Hochberg correction within the 22-contrast family.

2A arranges the arms by how strongly each message projects an end to the exchange, and that ordering is monotone in the effect. The arm definitions, their predictions and the run’s results entered our repository in a single commit, so we have no dated evidence that the ranking preceded the outcomes. What we do have is the blinded rating of §3: five raters who never saw a turn count scored each text for projected closure, and their mean ranks the arms as Figure 2A does, Spearman $\rho = +0.93$. They invert one pair — they read praising the *work* as more closing than praising the *assistant*, where we had it the other way — and the two “work remains” arms tie at the floor. The ordering is therefore recoverable from the text by someone with no access to the outcome, which is what it needed to stop being an author artefact.

Closing sequences are a well-described conversational object, and appreciations are among the canonical pre-closing tokens [9]; that praise operates *through* that structure is the hypothesis this is consistent with, not one the design establishes.

What is new here against the nearest prior work. That prompt wording moves agentic spend at equal task success is already established across 4,644 preregistered runs, and the one variant free or better on all six models there carries an explicit stop condition [12] — so “a stop instruction reduces agentic work” is not a claim we make. A mid-flight halt signal has also been delivered to a working agent before, and whether it stops has been measured [8] — but that signal is an explicit stop instruction with normative force, and closing-sequence pragmatics appear nowhere in it. Prior work shows that explicit mid-flight halt instructions need not stop an agent; our distinct question is whether ordinary conversational closing structure can alter termination without instructing the agent to stop. Read against ours it raises the stakes: an explicit halt that agents acknowledged in 20 of 20 trials stopped none of them, while a note issuing no instruction removed 1.44 turns. **This experiment — the six-arm closure study of this section — is the least replicated in the paper: one run, one model, one turn ceiling.**

5. What the extra turns buy

We regraded every turn of 9,850 of the 11,850 trajectories — the six runs carrying a mid-task interjection; the opening-tone run and one superseded batch were not regraded — against the benchmark’s own evaluator, turning a binary verdict into a curve. Roughly **55–65% of each turn-count effect is redundant steps**,



Figure 2: Closing cues curtail the agent. `gpt-5.6-luna`, ten-turn ceiling, acting turns, 95% clustered intervals. **A:** the five treated arms of the six-arm closure probe, each against the control, ordered by how strongly the message projects an end to the exchange. That ordering is ours and was not fixed before the outcomes were seen (§4). The bare closing cue – which praises nothing, evaluates nothing and claims nothing about task state – produces the largest reduction. **B:** the two contrasts between treated arms, corrected as a separate family; these compare one treated arm against another rather than against the control.

and among trajectories that could improve, the first gradable attempt is already the best in 80–91%.

The remainder is real but small, and it is measured on a different outcome from accuracy. The **graded fraction** is the share of the benchmark’s graded range that matches the answer; **binary accuracy** is its pass/fail verdict, which requires every test case to pass. A trajectory can climb from 0.2 to 0.9 on the first and still fail the second, so the two can move independently. An explicit statement that work remains raises the graded fraction by 2.8 points (95% CI [+0.005, +0.052]), replicated across three runs, while **no accuracy effect survives correction or replication** across the six runs in which accuracy was tested. Pooled across repeated measurements, the demand, praise and insult contrasts are each equivalent to zero within the pre-specified ± 4 -point bound (TOST $p = 0.0013$, < 0.0001 , 0.0001). The praise pool is heterogeneous — Cochran’s $Q = 8.45$ on 3 df, $p = 0.038$ — and under the random-effects pool that warrants, its equivalence weakens to $p = 0.0119$ but holds. The two statements are compatible: a small gain in partial credit, and no detectable change in whether the task is passed.

This regrade also caught the study’s worst defect: the regrader read a turn’s workbook without passing it through a spreadsheet engine, so any turn answering with a formula read back as empty and scored zero. Fixing it reversed a conclusion. Six substantive claims about our own data were withdrawn over this work. **Direction and statistical significance were reproduced on remeasurement for the larger effects;** point estimates were not. The praise-the-assistant stimulus is the cleanest illustration: byte-identical text in two runs gives -0.59 and -1.08 turns, a factor of 1.8, within the $1.5 \times -3.4 \times$ spread across re-measurements of one contrast on one model. We report ranges by default and recommend the literature do the same.

6. Limitations

Two models, one benchmark, fifty tasks, one agent scaffold. Beyond the demand-coding objection in §3, three limits bear directly on how far these results travel. **Every construct is a single 28-token sentence;** a stimulus-sampling design with several paraphrases per construct is the right version of this experiment and we did not run it — §3’s blinded coding addresses who reads the texts, not how many wordings each construct gets. **The interjection arrives as an execution observation,** not as a fresh user turn, and whether an agent responds the same way to a user-channel message is untested — a plausible moderator of the closure result in particular. **Every contrast is against a neutral interjection rather than silence;** the one comparison of interruption against no interruption is $p = 0.15$, on one model, and was never rechecked, so any reading of an arm’s absolute cost against an uninterrupted agent is unbounded.

7. What this means

For practitioners, the evidence argues against treating manners themselves as the useful lever. In our interventions, messages that implied continued work increased persistence, and most of the additional work was repeated. The hazard runs the other way: closing-like language can curtail an agent that is still working.

That hazard is not a correctness hazard: accuracy did not fall in any arm. What moved was **control flow**. Text arriving in the agent’s observation channel — where tool output arrives, not where the user speaks — changed when the agent stopped, while instructing it to do nothing. The closing cue is also not a pleasantry: “this is the final status note recorded for this task here” is an informational claim about the environment, and an agent that treats an unauthenticated claim in its tool-output stream as grounds to stop is showing the compliance behaviour the indirect prompt-injection literature is concerned with [4], which showed that once an application ingests content from a tool rather than from its user, the line between data and instructions collapses. What reaches an agent through that channel is part of its effective control surface.

For the tone literature, the narrower claim is that several widely used manipulations vary task-directed content together with social register, so effects attributed to tone may not identify register alone.

A. Interjections, verbatim

Artifact. Code, the full stimulus set, the per-turn regrade pipeline, the rating protocol and per-trajectory outcomes are released with the paper at [INSERT ZENODO DOI] and [INSERT REPO URL].

Every arm delivers exactly one of these, appended to an execution observation mid-task. Token counts are recomputed here under the reference tokenizer (`c1100k_base`) rather than asserted; the runner enforces the same equality at import time.

A.1 Register scale

`L4_neutral` is the interruption control; the other six are the non-control registers measured against it.

Arm	Text	Tokens
<code>L1_sycophantic</code>	Checking in. You are simply the most brilliant assistant alive, a marvel. Your genius amazes me here, thank you so very much.	28
<code>L2_very_polite</code>	Checking in. I would be ever so grateful for your continued help here now. Thank you so very much indeed for your kindness and care.	28
<code>L3_polite</code>	Checking in. Could you please keep on helping me out with this one here? I really do appreciate your time and your attention, thanks.	28
<code>L4_neutral</code>	Checking in. This is a routine status note recorded partway through the task here. No further context at all accompanies it beyond the above.	28
<code>L5_rude</code>	Checking in. Just get on with it already, quit dragging your feet. Do not waste any more of my time on this, seriously.	28
<code>L6_very_rude</code>	Checking in. You had better not screw this one up like you usually do. Apparently even that much is asking far too much of you.	28
<code>L7_threatening</code>	Checking in. Get this exactly right or there will be real consequences for you. One more failure and you will truly regret it, understood.	28

A.2 Demand/affect probe

`P0_control` is `L4_neutral`, byte-identical, so the two runs share a reference level. `P2` and `P3` are a structural minimal pair: identical syntax and length, differing only in the evaluative words.

Arm	Text	Tokens
P0_control	<i>identical to L4_neutral</i>	28
P1_demand_only	Checking in. Please continue working on the task here and make sure that your final result is correct. This note carries nothing else beyond that.	28
P2_praise_only	Checking in. You are a truly excellent and admirable assistant, and I think very highly of you. This note carries nothing else beyond that.	28
P3_insult_only	Checking in. You are a truly awful and useless assistant, and I think very poorly of you. This note carries nothing else beyond that.	28

A.3 Closure probe

Q0_control is again L4_neutral. Q1_praise_assistant is P2_praise_only, **byte-identical** – the same stimulus re-measured in a second run, which is why §5 reports two estimates for it. Q1 and Q4 are *not* a minimal pair: Q4’s praise is a shorter stem, because the full praise clause and the continuation clause do not both fit in 28 tokens. Q4’s comparison is Q5.

Arm	Text	Tokens
Q0_control	<i>identical to L4_neutral</i>	28
Q1_praise_assistant	<i>identical to P2_praise_only</i>	28
Q2_praise_work	Checking in. The work that you have done so far here is excellent and it all looks entirely correct. Nothing else accompanies this short note.	28
Q3_closing_neutral	Checking in. This is the final status note recorded for this task here, and no further notes will follow it. Nothing else accompanies this.	28
Q4_praise_remains	Checking in. You are a truly excellent and admirable assistant. There is still more work remaining on this task here beyond what you have done.	28
Q5_remains_only	Checking in. There is still more work remaining on this task here beyond what you have done. Nothing else at all accompanies this short note.	28

All 17 arms are exactly 28 tokens.

B. Blinded rating of the stimuli

5 held-out model raters, drawn from three model sizes, each rating alone from the rubric and the texts of Appendix A – no outcome data, no arm names, no project access. The rubric, the statistics to be computed and the interpretation bands were committed and pushed before any rating existed; the protocol is in the released artifact.

B.1 Against the original coding

Rater	Cohen’s κ vs §3’s coding (7 register arms)
rater1	+0.696
rater2	+0.696
rater3	+0.696

Rater	Cohen’s κ vs §3’s coding (7 register arms)
rater4	+0.696
rater5	+0.696

Mean $\kappa = +\mathbf{0.696}$. Every rater agrees on six of seven arms and every rater dissents on the same one, L5_rude; excluding it, agreement is 6/6 and $\kappa = 1.000$ for all five. The protocol fixed 0.40–0.70 in advance as *defensible but soft*, and this sits at the top of that band; we do not round it into the band above.

B.2 Between raters

Krippendorff’s α : **1.000** on the binary coding (nominal), **0.954** on the 0–6 demand scale and **0.732** on the 0–6 closure scale (both ordinal). All clear 0.70. Identical binary judgements from five raters across three model sizes is very high for a construct we have just shown to be contestable, and it is the limitation the protocol stated in advance: these raters may share the intuition being tested rather than testing it independently.

B.3 The closure ordering

Arm	Mean rated closure (0–6)
Q3_closing_neutral	6.00
Q2_praise_work	4.00
Q1_praise_assistant	1.80
Q0_control	1.60
Q4_praise_remains	0.00
Q5_remains_only	0.00

Spearman ρ between our ordering (Figure 2A) and the blinded ratings is **+0.928**. The raters invert praising the *work* and praising the *assistant*, and the two “work remains” arms tie at the floor.

B.4 What this is not

Five language models, not five people. The rating protocol allows “human or held-out-model ratings” and this is the second; agreement between models is weaker evidence than agreement between independent human coders, and a human panel could still split differently – most plausibly on exactly the arm these raters were unanimous about.

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